Ibrahim Joudah

Data Scientist, Statistician & Applied AI Specialist

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Meet me in 3 minutes

Professional Profile

- Data scientist and statistician with a **PhD in Statistics**. Has industry experience building reliable models, analytical tools and agentic AI systems that turn data into business decisions. Brings statistical rigour to machine learning, LLMs and GenAI. Advanced R user who also works in Python, SQL, Tableau and Power BI.

Featured Project

- **Agentic AI Monitoring and Decision Support Platform** 2026
  - Designed, built and run, end to end, an agentic AI platform (Python, SQL, LLMs, cloud server) that monitors **thousands of organisations' websites** and data feeds around the clock and flags relevant new postings within minutes.
  - Finds **4× the opportunities** that an LLM web search finds alone, in head-to-head tests.
  - A cascade of fast filters (rules in code, an XGBoost model and a cheap LLM screen) stops more than 95% of postings before any expensive AI evaluation, **cutting AI cost by about 95%** compared with evaluating everything with the most capable model.
  - A person reviews every shortlist and draft, and the AI's ratings are checked against blind human grades every week. Bayesian scheduling, automated tests and CI/CD keep it reliable.

Professional Experience

- **Statistical Consultant** Mar 2026 – present
  Pacific Analytics | Sydney, Australia
  - Develop DataSHIELD R packages for privacy-preserving federated analysis, including multiple imputation (MICE), survival analysis and regression modelling strategies (rms).
  - Adapt statistical methods to client-server limits, with automated tests and documentation.

- **Quantitative Data Lead** Aug 2025 – Jun 2026
  Altamir AI & Data Governance | Sydney, Australia
  - Reconciled the Australian Dental Association's records, which allowed it to release continuing professional development (CPD) certificates to dentists.
  - Led study design, data quality checks, analysis and reporting for external data governance and AI projects, with recommendations for senior stakeholders.
  - Supervised six UTS students through a three-month AI governance project for the media sector, including designing and analysing an **online survey of 153 adults**.

- **Teaching Associate** Feb 2023 – present
  Macquarie University | Sydney, Australia
  - Teach undergraduate and postgraduate statistics and machine learning in R and Excel.
  - Student evaluations consistently recognise the clarity and depth of my teaching.

- **Research Assistant** Dec 2022 – Jan 2023
  Environment Protection Authority Victoria | Australia
  - Contributed to the experimental design, analysis and reporting of a project estimating tyre numbers in stockpiles from 3D point cloud data.

- **Data Analyst Intern** Jan 2019 – Aug 2019
  Quant Data & Analytics | Riyadh, Saudi Arabia
  - Contributed to Fruits360, a SaaS analytics platform for restaurant sales and inventory, through data preparation, R integration, forecasting and Tableau dashboards.
  - Built automated anomaly detection that caught **100% of the data entry anomalies** in an audited sample of sales records, work that became a first-author paper in *IEEE Access* (2020).

Technical Skills

- **Languages & tools:** R (advanced), Python, SQL, Tableau, Power BI, Excel, Git
- **Applied AI:** Agentic AI, LLMs, GenAI, LLM evaluation, prompt design, NLP
- **Statistics & ML:** Statistical modelling, Bayesian modelling, regression, machine learning (XGBoost), deep learning, forecasting, anomaly detection, feature selection, experimental design
- **Software & MLOps:** Data pipelines, automation, production deployment (Linux, cloud), CI/CD, monitoring, R package development, DataSHIELD federated analysis

Selected Projects

- **Applied AI and predictive modelling for consumer surveys** 2026
  - Predicted survey qualification rates for categories with no labelled history, from LLM estimates, text embeddings and XGBoost (R, Python).
  - **65% lower prediction error** (RMSE) than the baseline of predicting the average rate, and 21% lower than the LLM estimates alone, in cross-validation on held-out categories.
  - A screening rule built on the model **clears 93% of qualifying questions** automatically, while still **catching 86% of low rate questions**.

- **StepAlytics: Interactive Dashboards on How Weather Shapes Daily Activity** 2026
  - Built interactive Tableau dashboards on a Python and SQL (DuckDB) analysis of a year of step tracker and weather data (the public version uses simulated activity data).
  - Quality checks traced an apparent rise in walking speed to a device change.

  Interactive Tableau dashboards | GitHub: analysis notes and report

- **Air-HOLP and AirScreen** 2024 – 2025
  - Feature screening method for high-dimensional data with strongly correlated predictors; first-author paper in *Statistics and Computing* and the AirScreen R package on CRAN.
  - Higher sure screening probability than Ridge-HOLP in **72% of 2,800 simulated settings**; on prostate cancer gene expression data, **multiple R of 0.85 against 0.80 (Ridge-HOLP) and 0.70 (SIS)**.

Education

- **Doctor of Philosophy in Statistics** 2022 – 2026
  Macquarie University | Sydney, Australia
  Thesis: *Feature Screening for Regression Models in Ultra-High Dimensional Settings*.
- **Master of Research in Statistics** 2021 – 2022
  Macquarie University | Sydney, Australia
- **Bachelor of Science in Actuarial Science** 2016 – 2020
  King Fahd University of Petroleum & Minerals | Dhahran, Saudi Arabia

Selected Publications

- **6** peer-reviewed papers (3 as lead author) | **100+** citations | **6** conference presentations
  - Air-HOLP: adaptive regularized feature screening for high-dimensional correlated data. *Statistics and Computing* (2025), first author.
  - Advancing student outcome predictions through generative adversarial networks. *Computers and Education: Artificial Intelligence* (2024).
  - Optimizing cancer classification in high-dimensional data by integrating generative AI with feature selection. *IEEE International Conference on Digital Health* (2026).

  Full publication list on Google Scholar

Awards & Recognition

- **Excellence in Postgraduate Research Award** 2025
  Statistical Society of Australia, NSW Branch.
- **International Research Training Program (iRTP) Scholarship** 2022 – 2025
  Australian Government, at Macquarie University, **valued at A$182,616**.